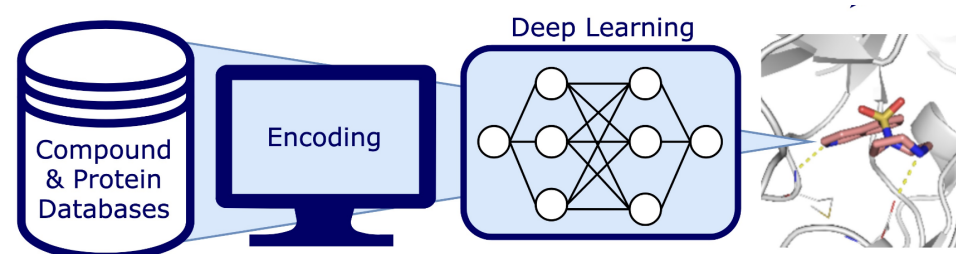


Data Driven Drug Design

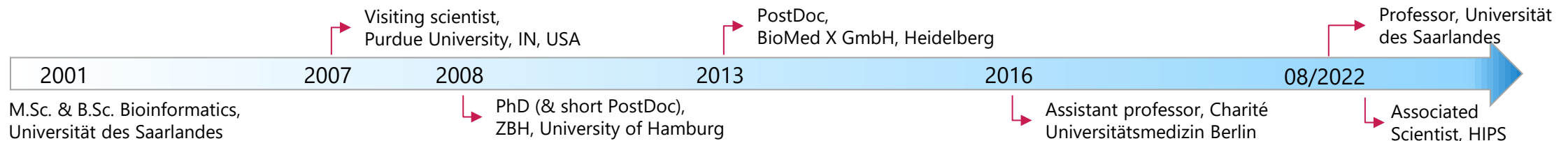
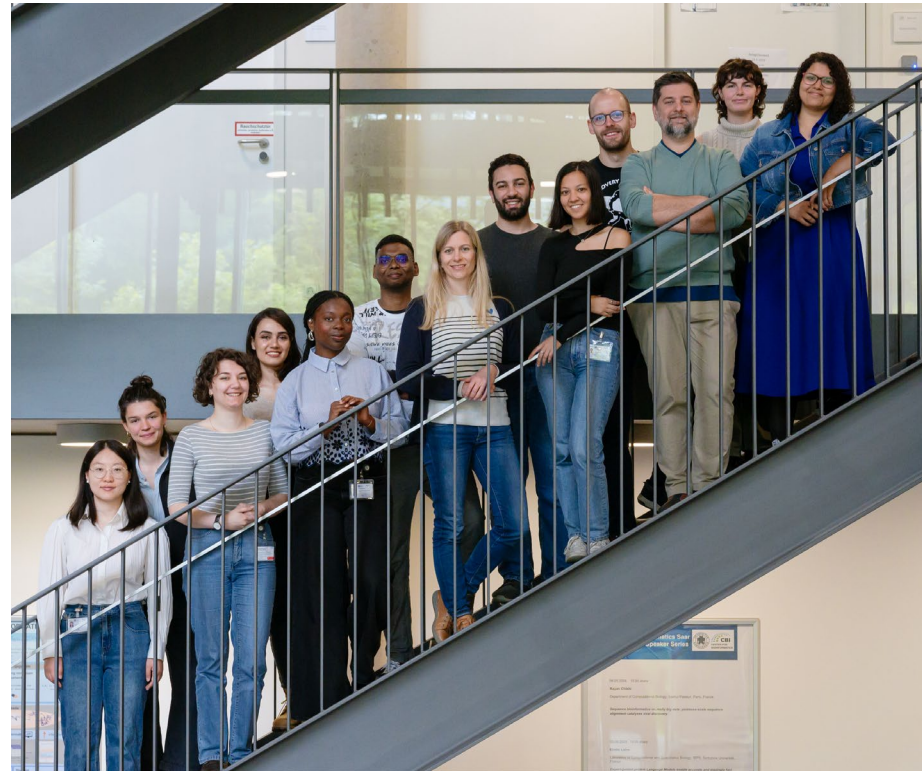
Machine Learning with Python

Prof. Dr. Andrea Volkamer
Data Driven Drug Design Group



Data Driven Drug Design

- Method development and application in computer-aided drug design (CADD) and predictive toxicology
- Focus on integration of structural and chemical information in machine learning
- Application in cancer research and antiinfectives
- <https://volkamerlab.org/>



Who we are ...

Andrea Volkamer
Lecture



Katharina Buchthal
Tutorium

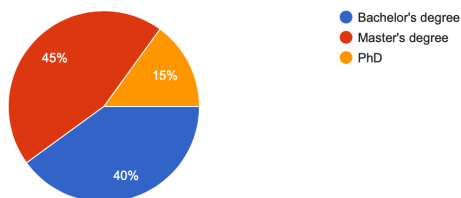


Who you are ...

What is the highest degree you have completed?

20 Antworten

 Diagramm kopieren



Workshop Participants' Areas of Expertise

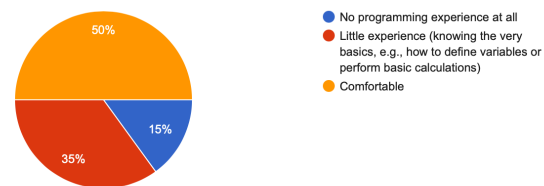


Python and data preparation

What is your level in programming, specifically with Python?

20 Antworten

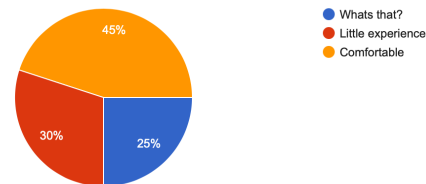
 Diagramm kopieren



How familiar are you with Pandas or Numpy?

20 Antworten

 Diagramm kopieren

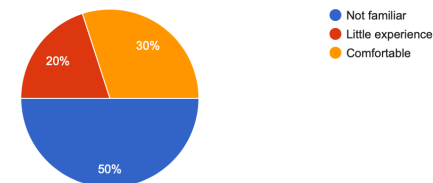


Machine learning

What is your level of expertise with machine learning (ML) in drug discovery?

20 Antworten

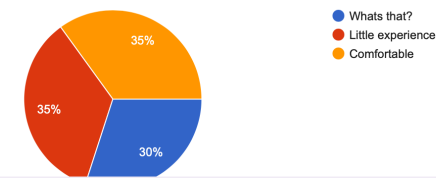
 Diagramm kopieren



How familiar are you with scikit-learn?

20 Antworten

 Diagramm kopieren

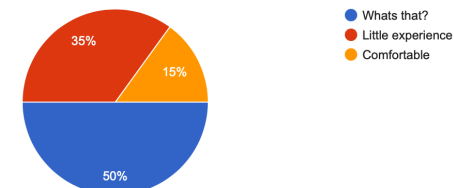


Deep learning

How familiar are you with PyTorch?

20 Antworten

 Diagramm kopieren



Machine Learning (ML) for CADD in Python

- **From molecular data to models**

Learn data preprocessing, molecular representations, and their role in ML

- **Machine learning essentials**

Explore unsupervised (clustering) & supervised methods (RF, SVM) and performance metrics

- **Deep learning applications**

Understand molecular encodings and modern model architectures

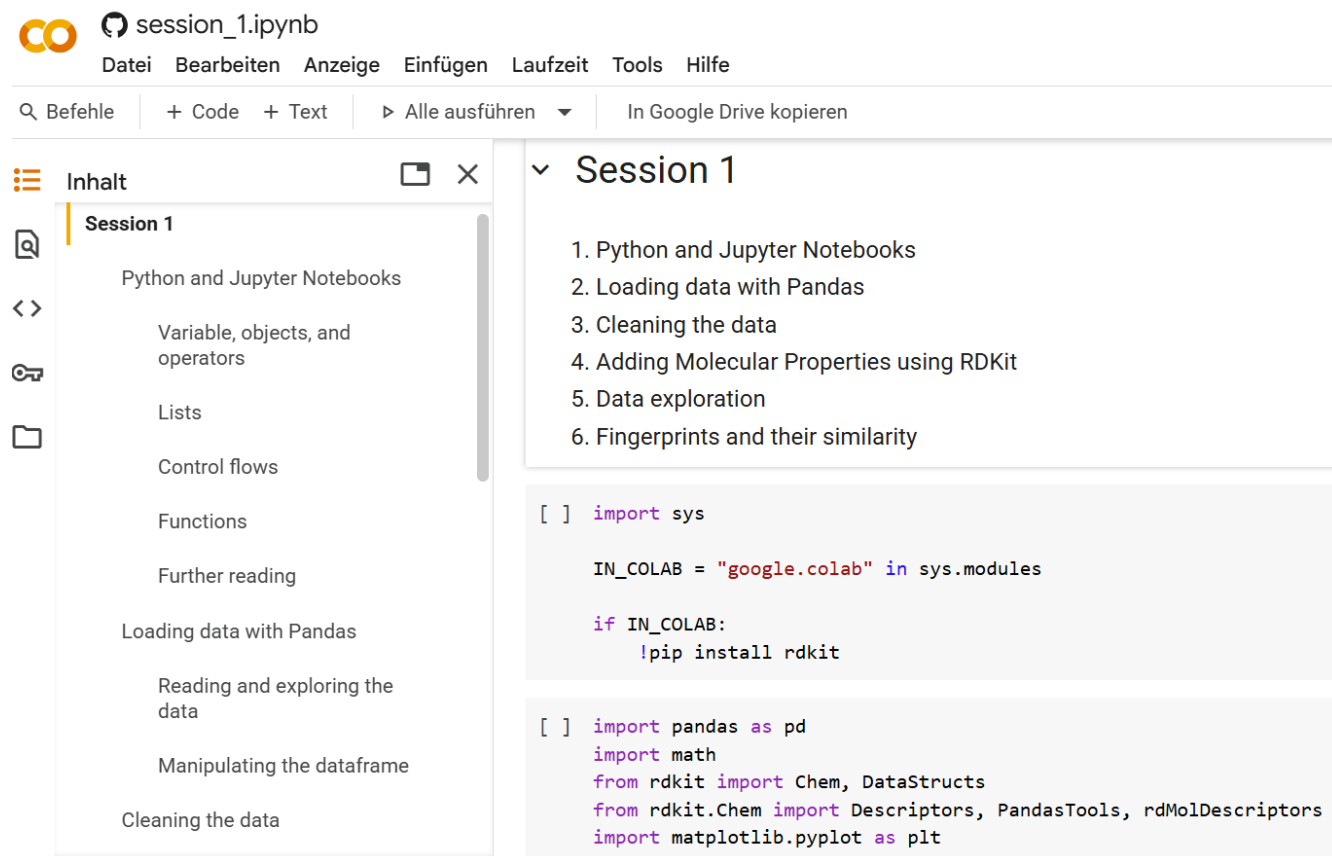
Workshop Structure

1.5h per session

- Lecture ~40 min
- Hands-on part ~50 min
 - Guided Jupyter tutorial ~25 min
 - Own exercises ~25 min
 - Executed on colab
 - No installation needed!

All material available at:

https://github.com/volkamerlab/cic_summerschool_2025



The screenshot shows a Jupyter Notebook interface for a file named 'session_1.ipynb'. The top menu bar includes 'Datei', 'Bearbeiten', 'Anzeige', 'Einfügen', 'Laufzeit', 'Tools', and 'Hilfe'. Below the menu is a toolbar with 'Befehle', '+ Code', '+ Text', 'Alle ausführen', and 'In Google Drive kopieren'. On the left, a sidebar titled 'Inhalt' displays a table of contents for 'Session 1' with items like 'Python and Jupyter Notebooks', 'Variable, objects, and operators', 'Lists', 'Control flows', 'Functions', 'Further reading', 'Loading data with Pandas', 'Reading and exploring the data', 'Manipulating the dataframe', and 'Cleaning the data'. The main area shows 'Session 1' with a list of topics: 1. Python and Jupyter Notebooks, 2. Loading data with Pandas, 3. Cleaning the data, 4. Adding Molecular Properties using RDKit, 5. Data exploration, and 6. Fingerprints and their similarity. Below the list are two code cells. The first cell contains code to set the environment for Google Colab and install RDKit. The second cell contains code to import pandas, math, and various modules from RDKit and matplotlib.

```
[ ] import sys

IN_COLAB = "google.colab" in sys.modules

if IN_COLAB:
    !pip install rdkit

[ ] import pandas as pd
import math
from rdkit import Chem, DataStructs
from rdkit.Chem import Descriptors, PandasTools, rdMolDescriptors
import matplotlib.pyplot as plt
```

What we Will do in More Detail ...

Session 1 - 8:30 - 10:00 am

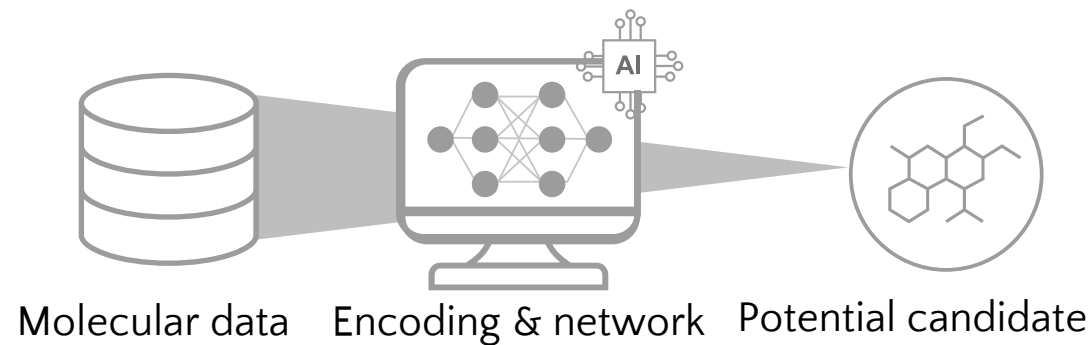
- CADD, Artificial Intelligence (AI) and Machine Learning (ML)
- The importance of data quality for AI/ML: Data preprocessing and analysis
- Molecular representations: Fingerprints and similarities

Session 2 - 10:00 - 11:30 am (+ exercise buffer after lunch)

- Introduction to machine learning (ML)
- (Un)supervised learning: First models, e.g., RF, and SVM
- Performance evaluation

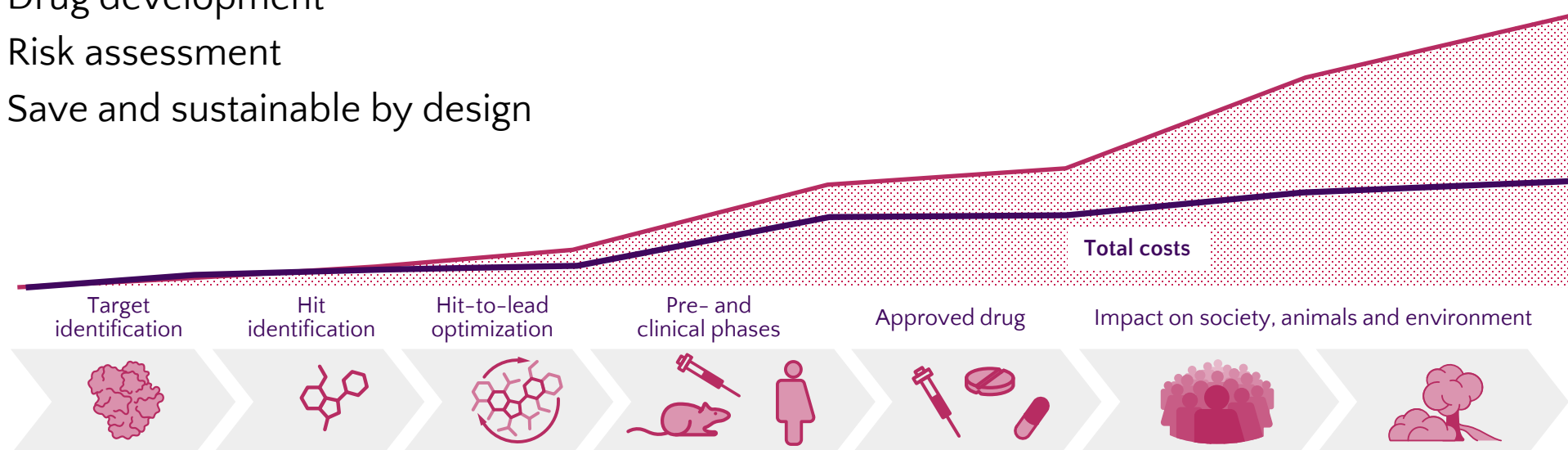
Session 3 - 13:00 - 2:30 pm

- Deep learning
- Neuronal networks
- Molecular encodings and architectures



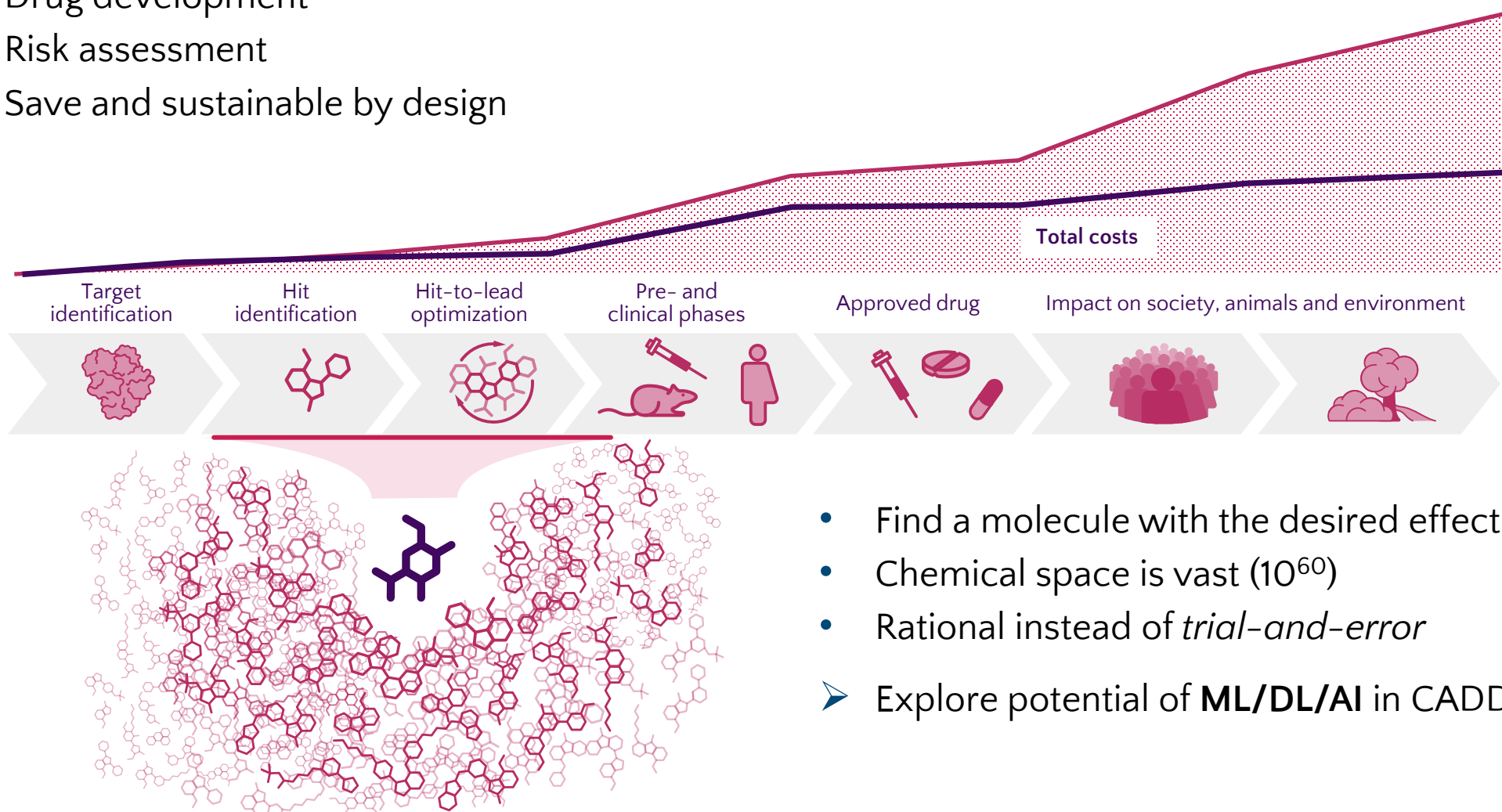
Computer-Accelerated Prioritization of Molecules

- Drug development
- Risk assessment
- Save and sustainable by design



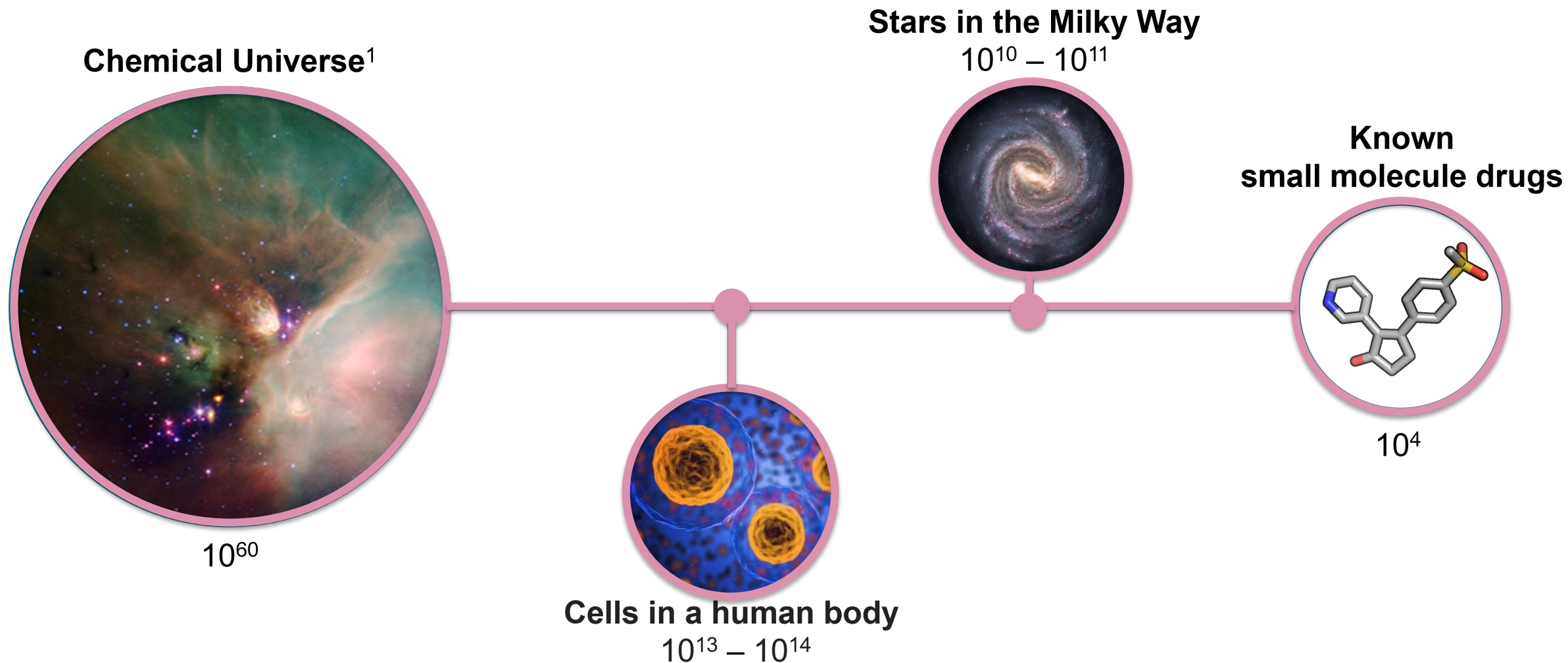
Computer-Accelerated Prioritization of Molecules

- Drug development
- Risk assessment
- Save and sustainable by design



- Find a molecule with the desired effect
 - Chemical space is vast (10^{60})
 - Rational instead of *trial-and-error*
- Explore potential of **ML/DL/AI** in CADD

Drug Discovery is a Chemical Space Odyssey

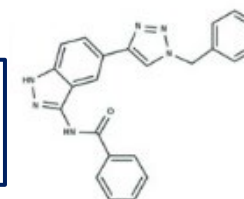
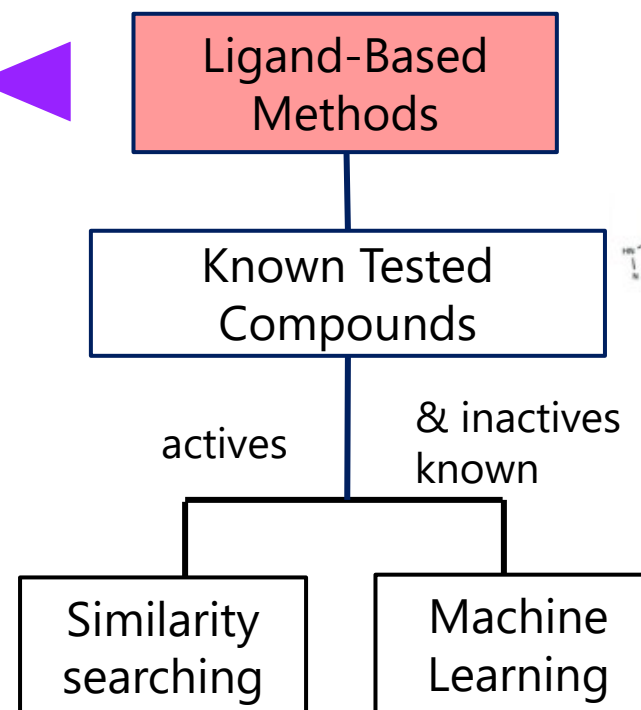
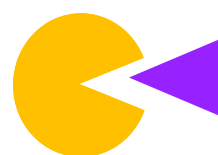
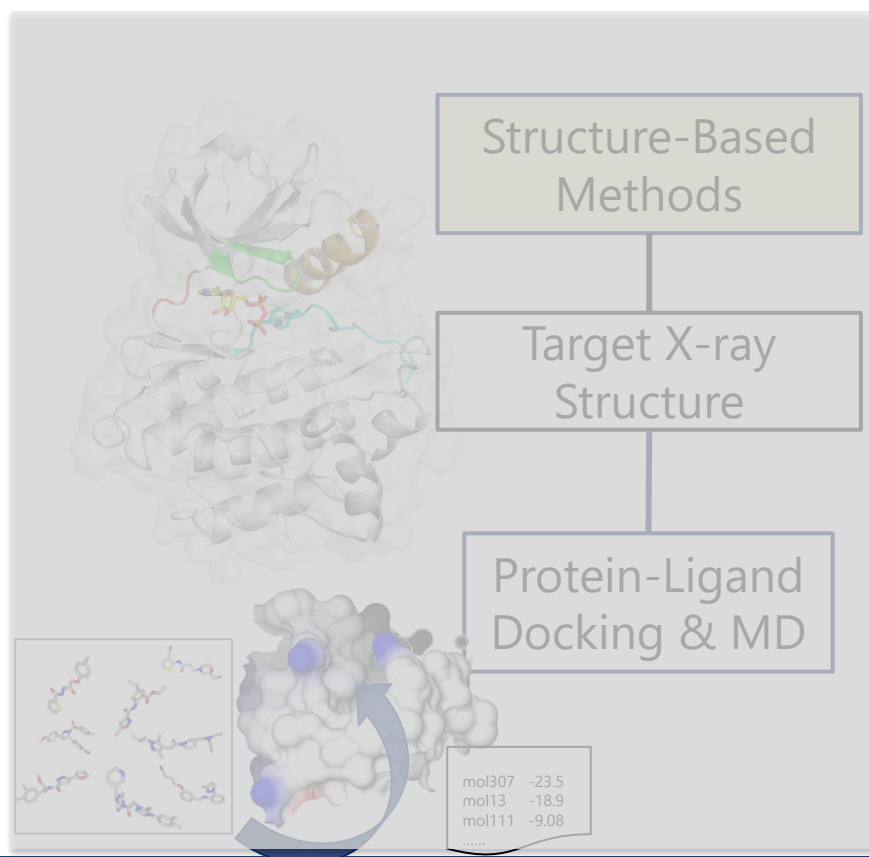


¹Walters et al. *Drug Discovery Today* 1998; Ertl. *J. Chem. Inf. Comp. Sc.* 2002.

Slide adapted from F. Grisoni

Finding New Drug Candidates – Virtual Screening

- *In silico* techniques used to search large compound data bases to select a smaller number of compounds for biological testing
- Chosen method depends on available data



Ligand-Based Methods: Principle of Molecular Similarity

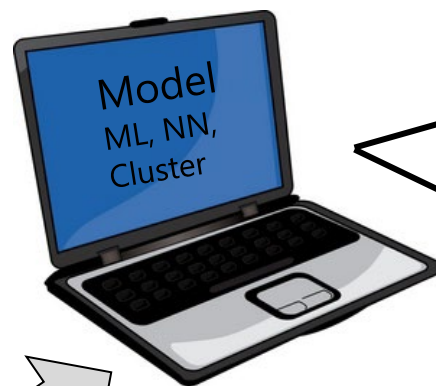
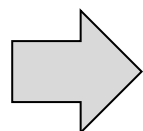
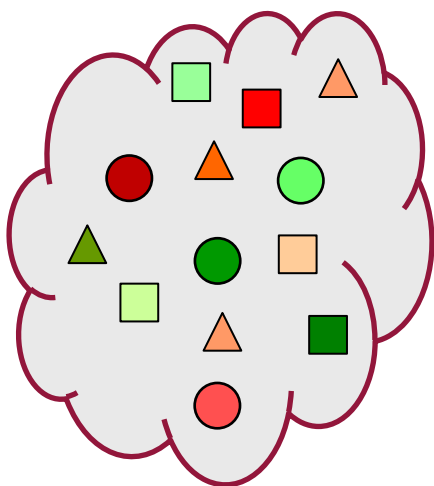
Assumption: Structurally similar molecules have similar properties and, thus, similar biological activity

- Given a set of molecules with known binding affinity, find further molecules

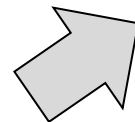
Library of known active and inactive compounds against a target of interest

Requirements:

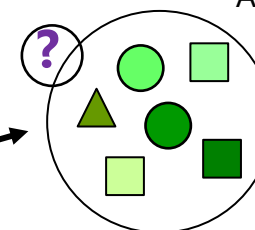
- Machine representation of the molecules
- Quantitative measure of similarity between the molecular descriptors



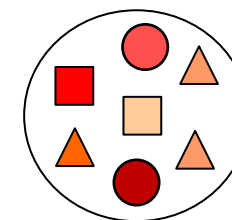
New query compound



Active



Inactive

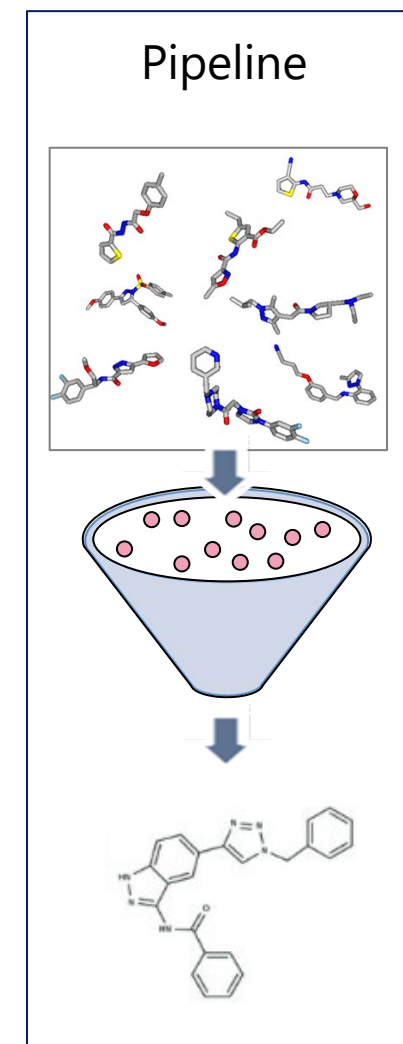
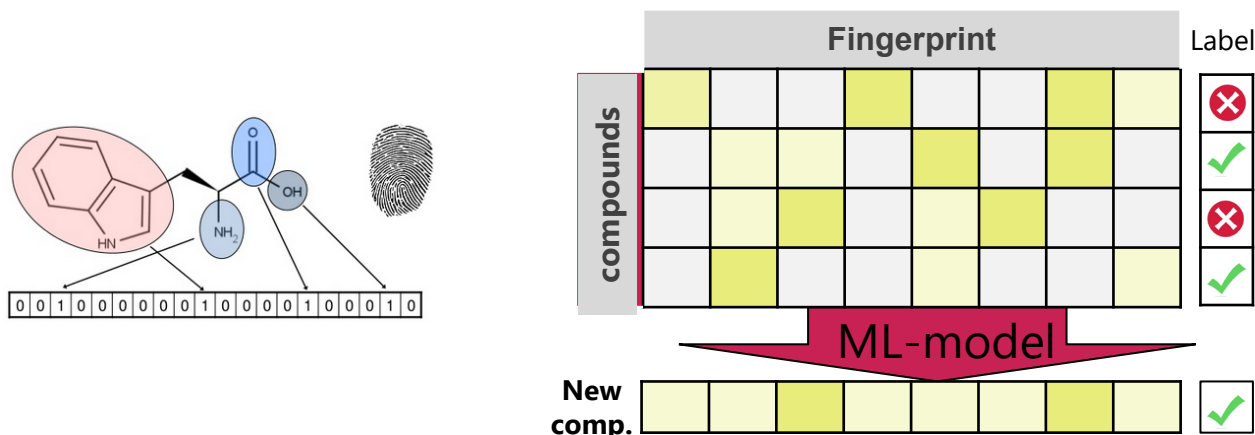


- Learn the criteria that are important to distinguish between different outcomes

Ligand-Based Methods: Similarity Search and Machine Learning

Given a set of molecules with known binding affinity to a specific target ...

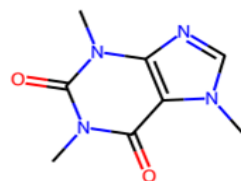
- Similarity Search: Find similar molecules
 - Transform molecules into fingerprints
 - Tanimoto similarity between each pair of fingerprints
- Quantitative structure activity relationships (QSAR) [machine learning]
 - Derive/learn the criteria to distinguish between different outcomes
 - Relate a set of *predictor* variables (X) to the *response* variable (Y)



Key Components of Machine Learning in Drug Discovery (or Anything Else)

(Training) Data

0.0009	0.004	0.0008				26.7	4.2		0.232		
0.001	0.004	0.0006				36.2	5.2		0.179		
0.001	0.736	0.0076									
0.001	0.004	0.001				27.5	3.5		0.262		
0.001	0.017	0.0004	<1.0			30.5	3.8		1.4		
0.0015	0.041	0.0415	38.3						12		
0.002	0.02	0.028	385.2			30	2.5		0.584		
0.001	0.0005	0.004	386.7			24.4	3.5		3.1		
0.0006	0.0006	0.0026	326.5	34	5.2	70.4	3.2	496	9.15	0.25	0.291
0.001	0.0145	0.001	4.1			26.2	3.5		1.3		
0.001	3.259	0.0048	5.3						2.2		
0.001	0.007	0.0006	375.0			32.5	3.4		0.395		
0.0016	0.73133	0.0003	6.9	186.3	4.3	72.1	3.8	7.3	14.45	1	7.4
0.001	0.007	0.0008	337.0			26.8	3.9		5		
0.0006	0.0075	0.0008	56.2			5.9	4.3		0.218		
0.001	0.0095	0.002							3.1		
0.0025	16.885	0.121									
0.0028	0.054687	0.0005	42.2	4		27.1	4.5	28.4	2.4	2	0.090
0.0007	0.006	0.004				21.6	36.2		6		
0.001	0.001	0.0003				53.0	5.3		7.5		
0.001	0.207	0.026				0.491	6.2		36		
0.001	0.2715	0.0041							0.397		
0.001	0.3515	0.006							0.363		



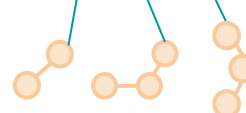
Representation

SMILES

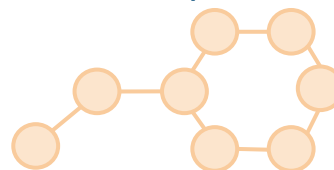
CN1C=NC2=C1C(=O)N(C(=O)N2C)C

Fingerprints

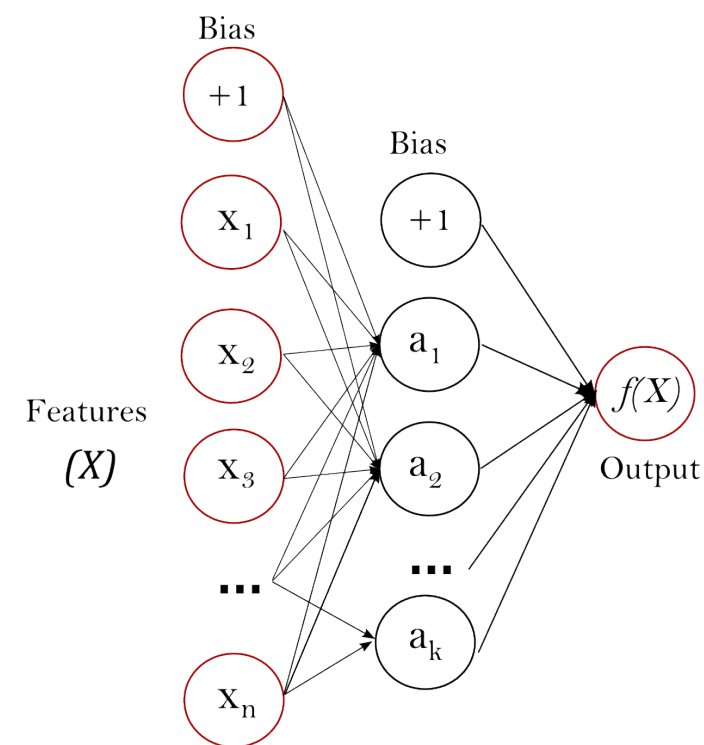
0 0 1 0 1 0 1 0 0



Graph



Architecture



What we will do in detail ...

Session 1

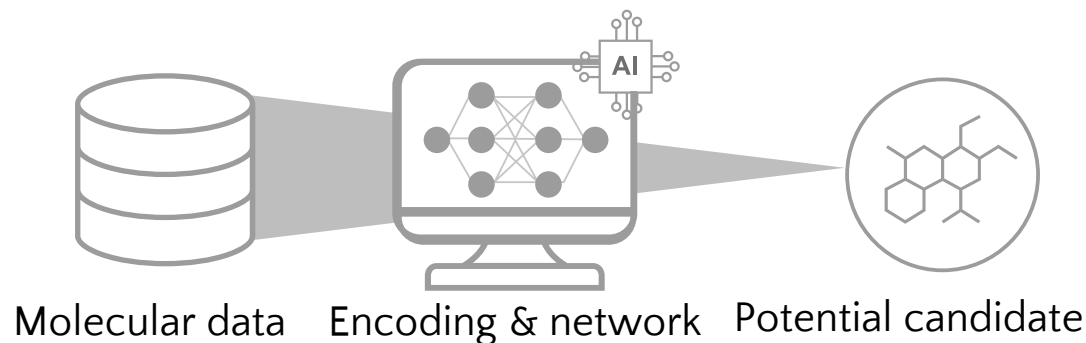
- CADD, Artificial Intelligence (AI) and Machine Learning (ML)
- **The importance of data quality for AI/ML: Data preprocessing and analysis**
- Molecular representations: Fingerprints and similarities

Session 2

- Introduction machine learning (ML)
- (Un)supervised learning: RF, SVM, ...
- Performance evaluation

Session 3

- Deep learning
- Neuronal networks
- Molecular encodings and architectures



AI (R)Evolution – The Transforming Role of Data

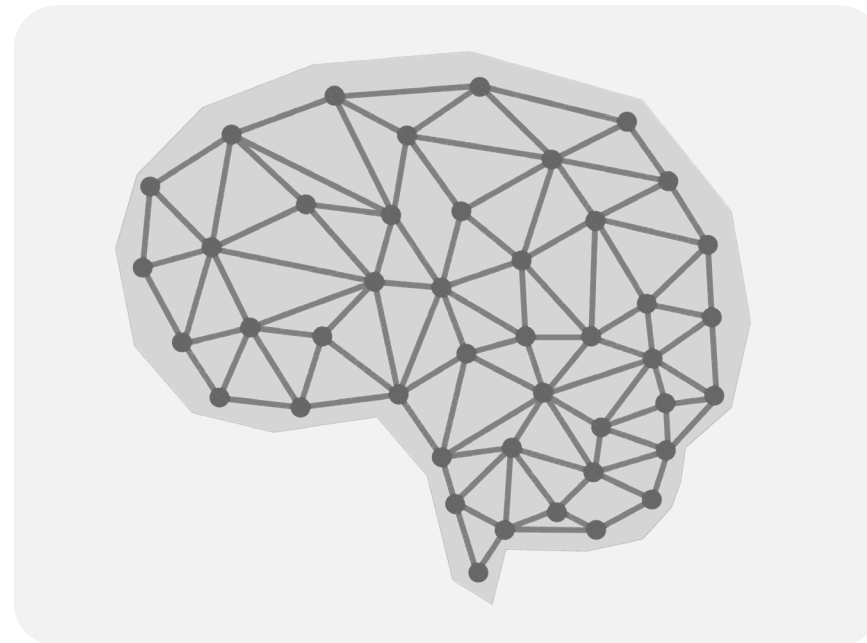
- Good performance: **well-defined problem** and **many examples** for training

How can I help you?



Generated by DALL-E

Artificial neural network



Structure prediction

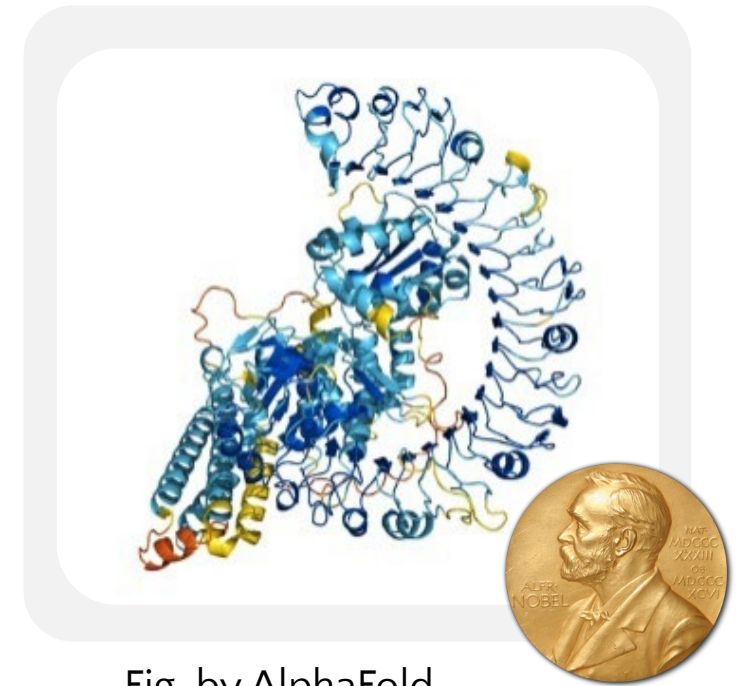


Fig. by AlphaFold

(Ultra)Large amounts of data

- Pharmaceutical data is miniscule compared to many other fields

Responses are definitive

Samples are independently distributed

- No cases where the same example falls into 2 different categories

Samples are identically distributed

- Equal distributions of positive and negative examples

Training data is representative of what is being predicted

Data is sparse

- Rarely have a complete data matrix

Data is truncated

- Many assay values report as "<1" or ">30"
- Difficult to know the true value

Data has a limited dynamic range

- Often spans only 2 or 3 logs
- Small dynamic range combined with experimental error makes significant correlations difficult

Even data from the "same" assay can be heterogeneous

- PK data measured with different doses, formulations
- Response can vary with operator, equipment, lab

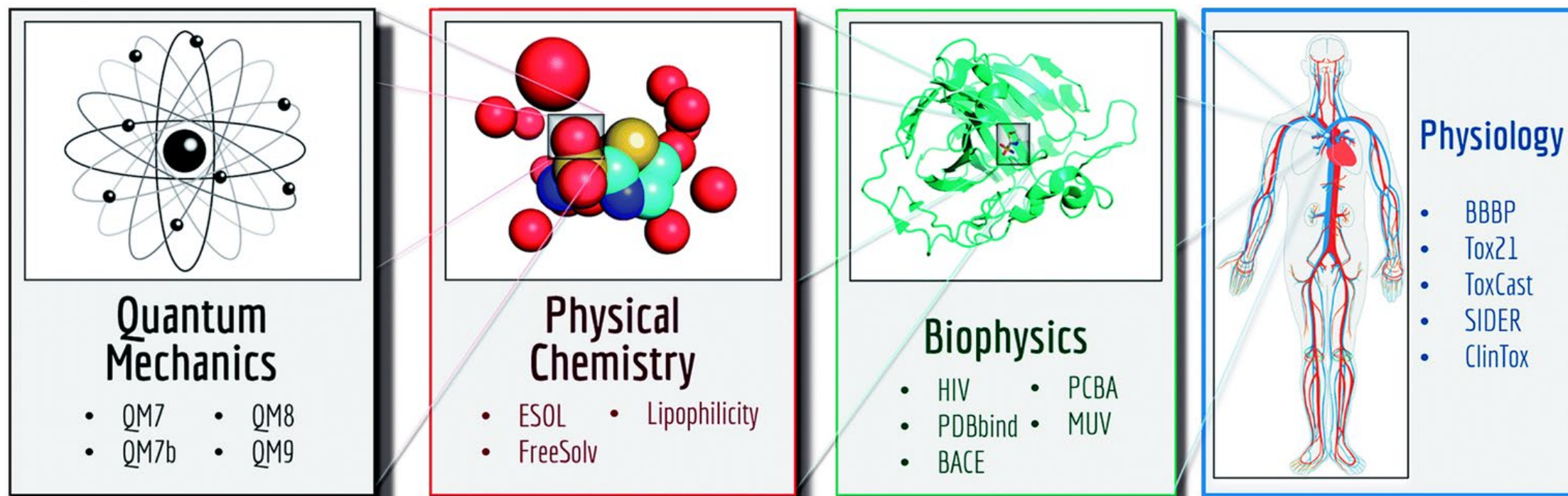
Data covers a limited chemical space

- Even global models can be local

What kind of data sets do we have for training ML models?

Small Molecule Benchmark Datasets: MoleculeNet

Common molecule benchmarks for molecular property prediction:
16 datasets divided into 4 categories.



<https://moleculenet.org/>

Small Molecule Benchmark Datasets: MoleculeNet

Quantum Mechanics data: QM9 → regression tasks

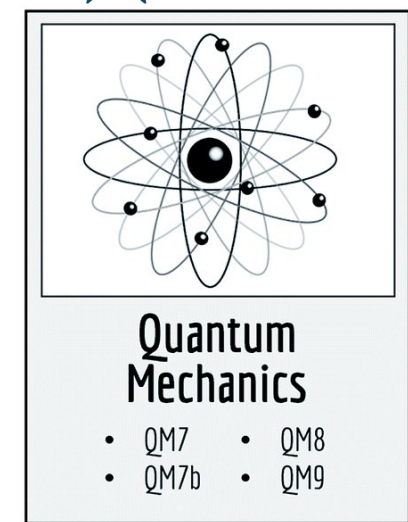
- -134k molecules (SMILES, 3D) with ≤ 9 heavy atoms
- 12 properties: Geometric, energetic, electronic, and thermodynamic properties of DFT-modelled small molecules
- Commonly used due to its comparably large size

Tox21 → classification

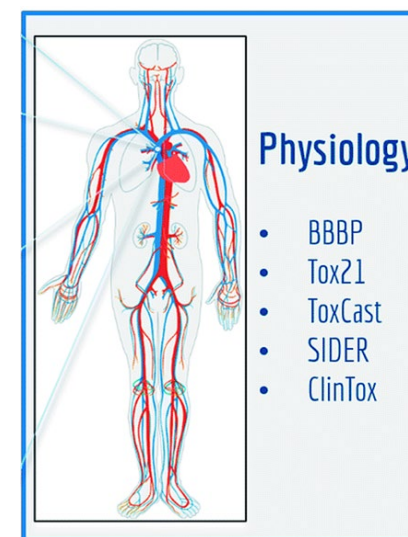
- Toxicity data with molecules measured against toxicity-relevant endpoints
- -8k molecules
- 12 tasks, here biological targets, including nuclear receptors and stress response pathways

CAREFUL! See Pat's BlogPost

<https://practicalcheminformatics.blogspot.com/2023/08/we-need-better-benchmarks-for-machine.html>



<https://moleculenet.org/>



Benchmark Data Sets: Therapeutic Data Commons (TDC)

Artificial intelligence foundation for therapeutic science

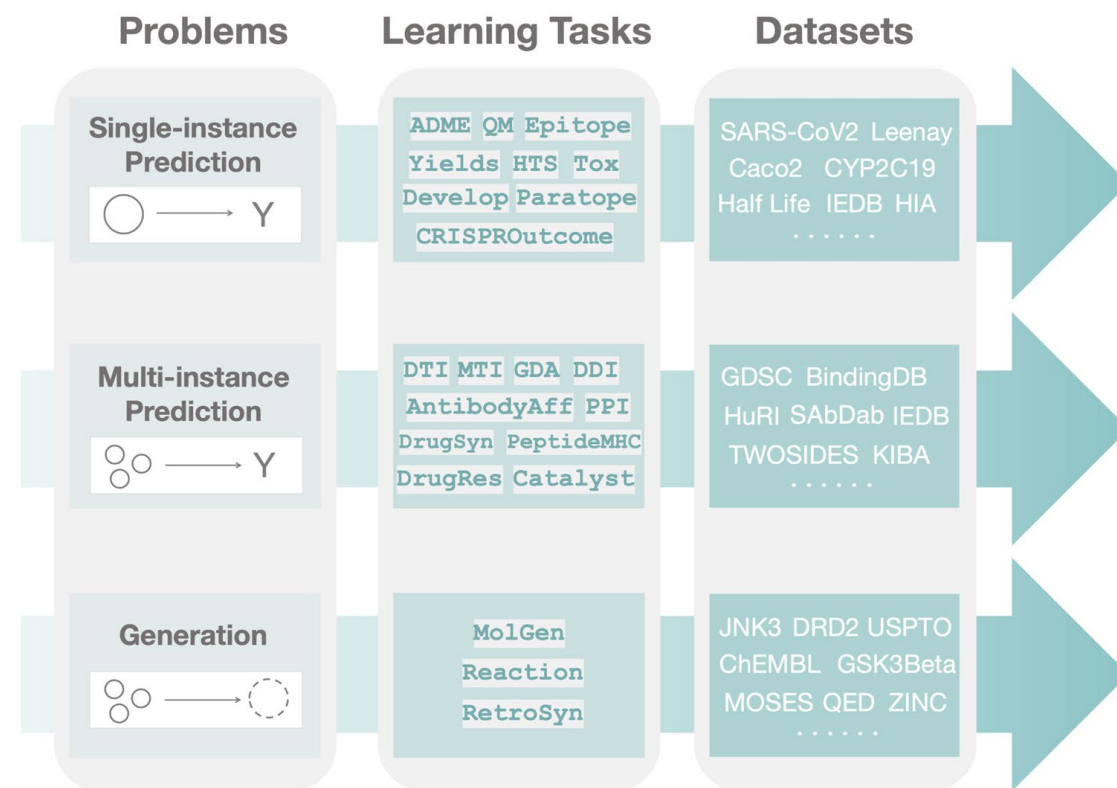
Artificial intelligence (AI) is poised to transform therapeutic science. Therapeutics Data Commons is an initiative to access and evaluate AI capability across therapeutic modalities and stages of discovery, establishing a foundation for understanding which AI methods are most suitable and why.

Kexin Huang, Tianfan Fu, Wenhao Gao, Yue Zhao, Yusuf Roohani, Jure Leskovec, Connor W. Coley, Cao Xiao, Jimeng Sun and Marinka Zitnik

Nat Chem Biol (2022). <https://doi.org/10.1038/s41589-022-01131-2>

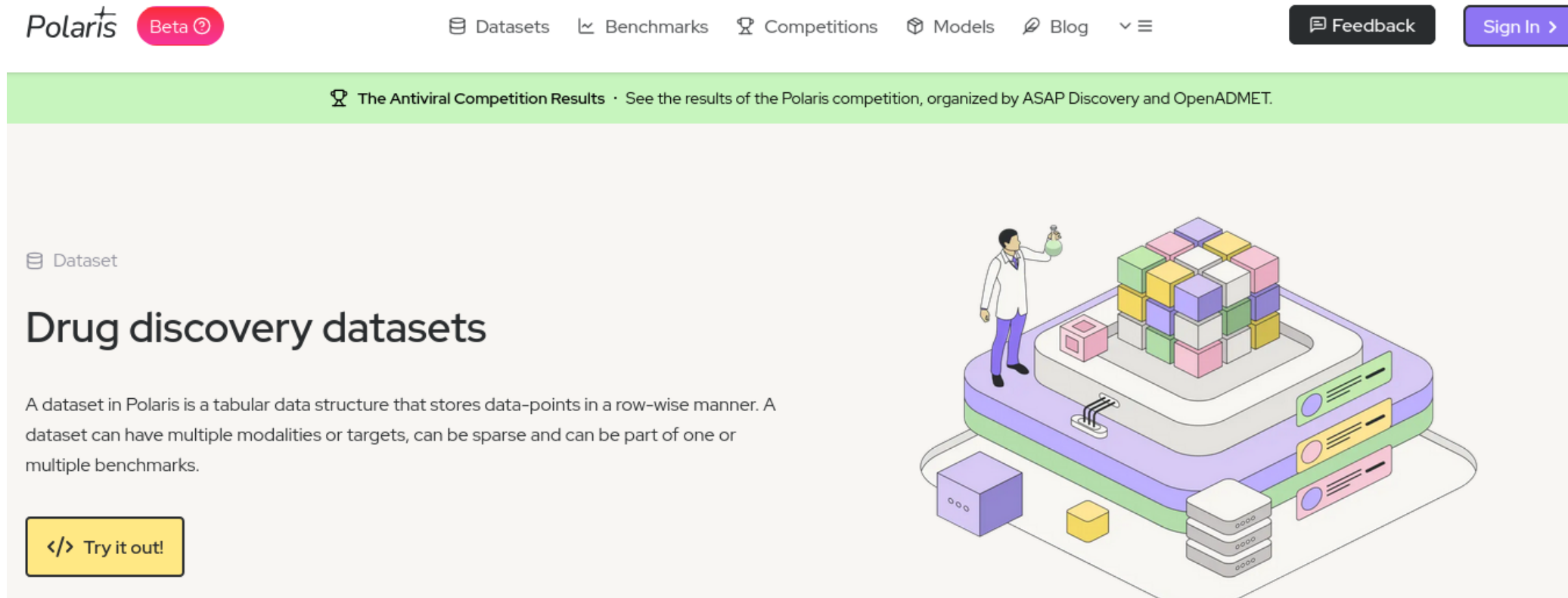
CAREFUL! See Pat's BlogPost

<https://practicalcheminformatics.blogspot.com/2023/08/we-need-better-benchmarks-for-machine.html>

 Check for updates[comment](#)

Single Origin and Well-Curated Datasets - Polaris Initiative

<https://polarishub.io/>



The screenshot shows the Polaris website interface. At the top, the Polaris logo is followed by a 'Beta' badge. Navigation links include 'Datasets', 'Benchmarks', 'Competitions', 'Models', and 'Blog'. There are also 'Feedback' and 'Sign In' buttons. A green banner highlights 'The Antiviral Competition Results'. The main section is titled 'Dataset' and 'Drug discovery datasets'. It includes a description: 'A dataset in Polaris is a tabular data structure that stores data-points in a row-wise manner. A dataset can have multiple modalities or targets, can be sparse and can be part of one or multiple benchmarks.' Below this is a 'Try it out!' button. To the right is a 3D illustration of a scientist standing next to a large, colorful, multi-layered cube structure, symbolizing data complexity and discovery.

Polaris⁺ Beta

Datasets Benchmarks Competitions Models Blog

Feedback Sign In

The Antiviral Competition Results · See the results of the Polaris competition, organized by ASAP Discovery and OpenADMET.

Dataset

Drug discovery datasets

A dataset in Polaris is a tabular data structure that stores data-points in a row-wise manner. A dataset can have multiple modalities or targets, can be sparse and can be part of one or multiple benchmarks.

</> Try it out!

Inspired by (beneath others) Pat's BlogPost

<https://practicalcheminformatics.blogspot.com/2023/08/we-need-better-benchmarks-for-machine.html>

Small Molecule Datasets (unlabeled): ZINC and GuacaMol

ZINC (<https://pubs.acs.org/doi/full/10.1021/acs.jcim.5b00559>)

- Ultralarge collection of purchasable, “drug-like” molecules without target value
- ZINC15: ~120M molecules
- ZINC20: ~1.4 billion molecules,

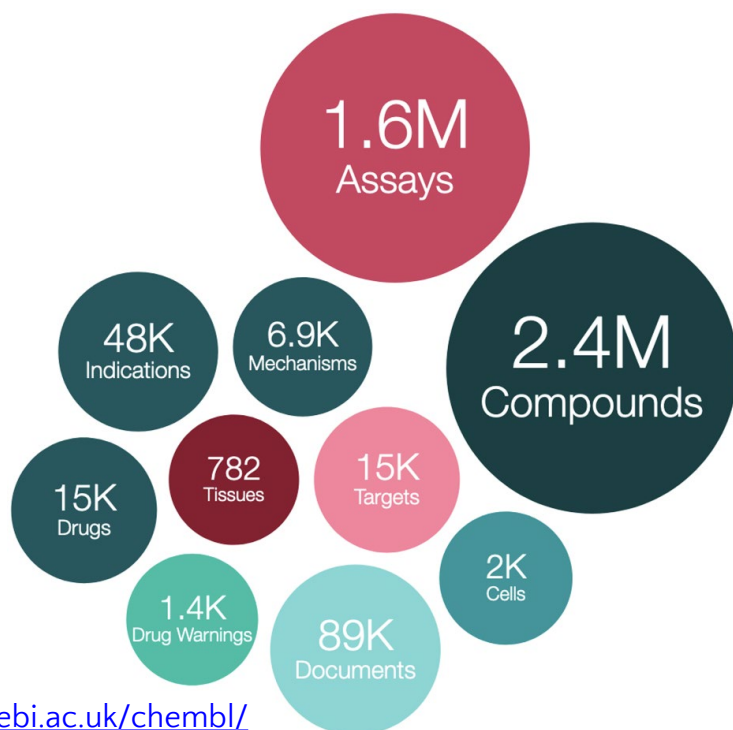
GuacaMol (<https://pubs.acs.org/doi/10.1021/acs.jcim.8b00839>)

- Large collection of molecules from ChEMBL 24 without target value
- ~1.5M molecules
- Published training, test and validation set → for benchmarking

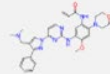
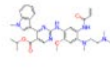
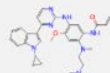
→ Commonly used benchmark for pretraining (LLMs) and generative models

Large Pool of Data Available – ChEMBL Database

- Manually curated – publicly available - chemical database of bioactive molecules
 - Contains drugs, drug-like small molecules and their targets
- Database for collecting binding affinities

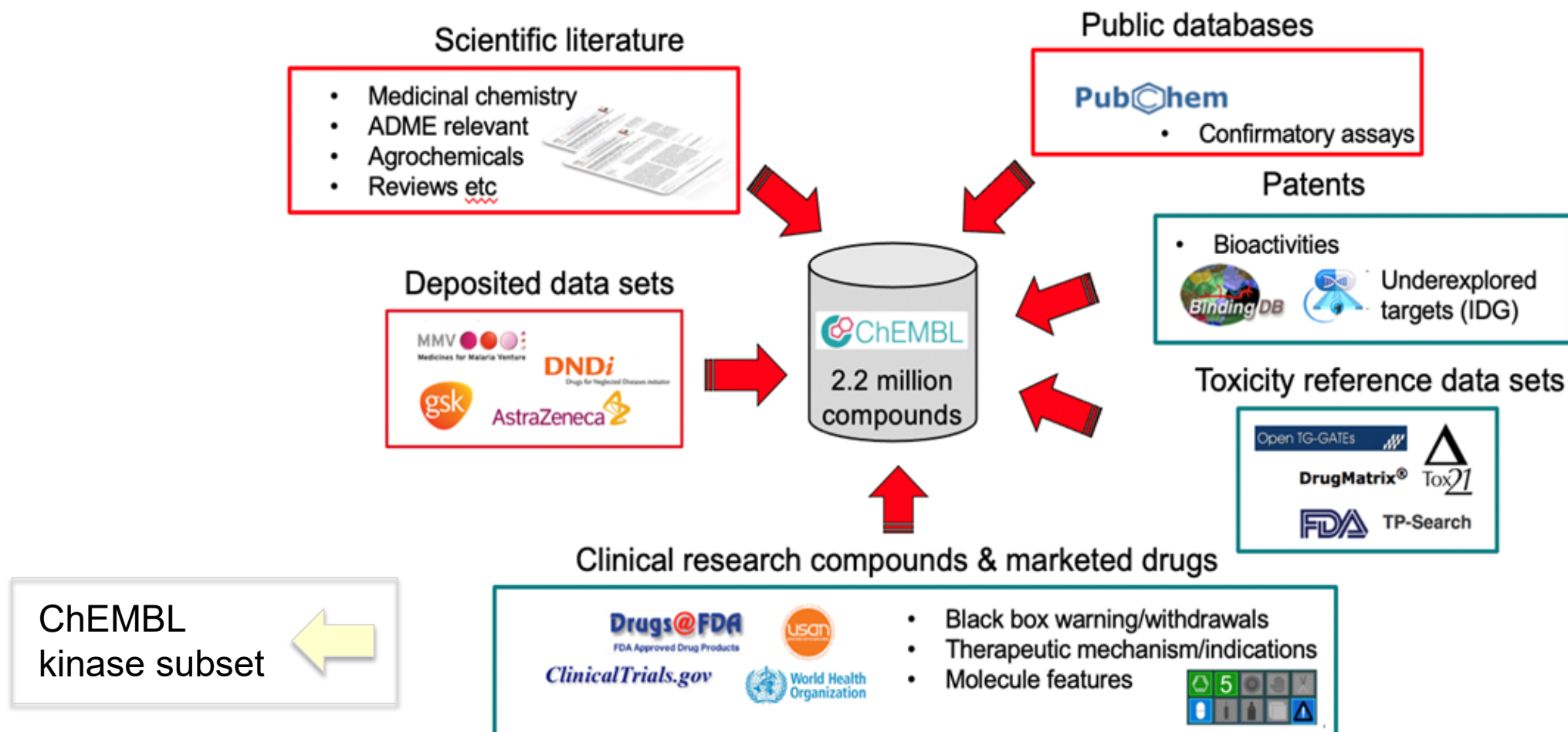


<https://www.ebi.ac.uk/chembl/>

ChEMBL ID	Search Hit	Name	Synonyms	Type	Max Phase	Molecular Weight	Targets	Bioactivities
 CHEMBL4558324		LAZERTINIB	C-18112003-G, GNS1480, GNS-1480, JNJ-73841937-AAA, Lazertinib, Yh25448, YH25448, YH-25448	Small molecule	3	554.66	12 By Type:	44 By Std. Type:
 CHEMBL4650319		MOBOCERTINIB	AP32788, AP-32788, Mobocertinib, Tak-788, TAK-788	Small molecule	4	585.71	14 By Type:	55 By Std. Type:
 CHEMBL4761468		AUMOLERTINIB	Almonertinib, Ameile, Amerol, Aumolertinib, Egfr t790m inhibitor hs-10296, EQ143, EQ-143, HS-10206, Hs 10296, Hs-10296, HS-10296	Small molecule	3	525.66	5 By Type:	23 By Std. Type:

ChEMBL - Where Does it Come From?

(Public) datasets are never perfect.

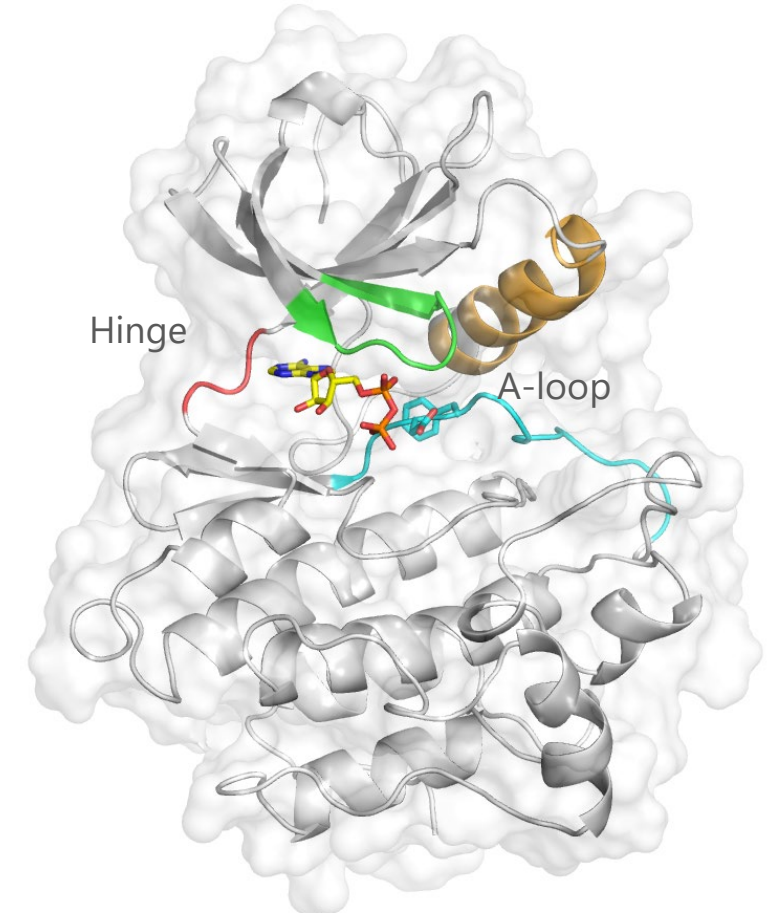


Kinases in a nutshell

- Established drug targets for cancer and inflammatory diseases
- Human kinome consists of nearly 518 protein kinases
- > 6000 human kinase X-ray structures available in the PDB
- Roughly 70 FDA approved small kinase inhibitors on the market
- Most of the approved drugs bind in the ATP-binding pocket

Problems:

Resistance to current drugs & broad pharmacology profile due to high-similarity of the ATP-binding pocket (major target)

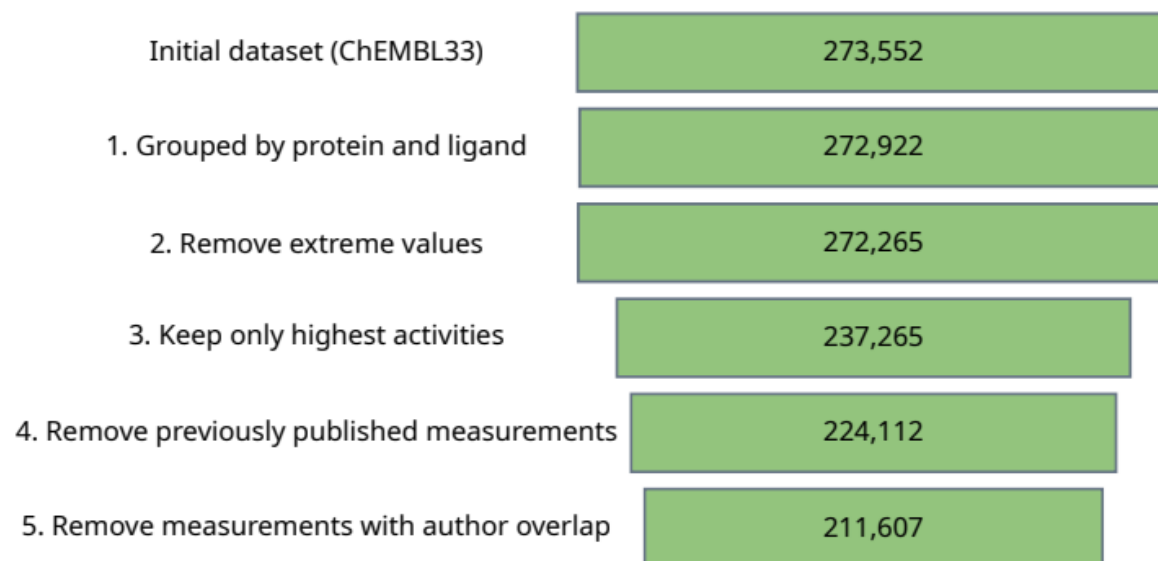


3D structure of a protein kinase and its ATP-binding site.

How to Preprocess such Data Properly?

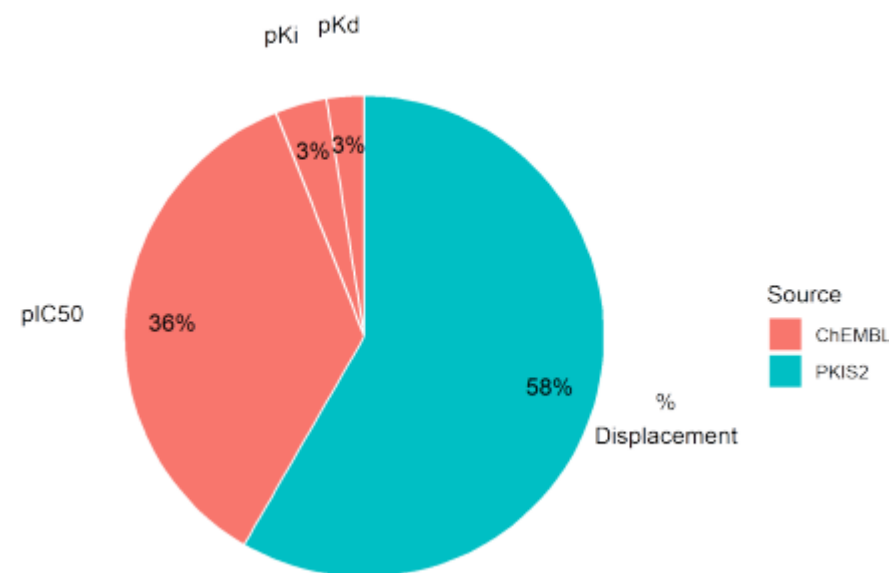
- Automated curation pipeline to reduce errors and to increase reliability

see Kramer, et al. *JMedChem* **2012**; 55(11):5165–5173



López-Ríos de Castro, et al., *bioRxiv*, **2024**
<https://doi.org/10.1101/2024.09.10.612176>

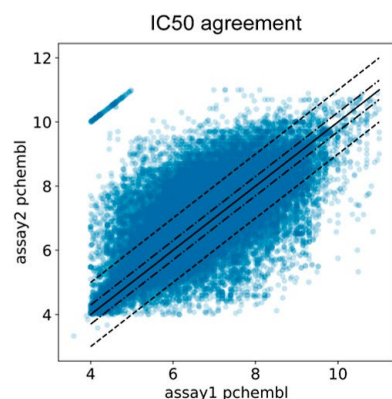
- Bioactivity assay measurement classes vary by and within data set



→ ChEMBL
EGFR IC50 subset

Different Assays – Different Results

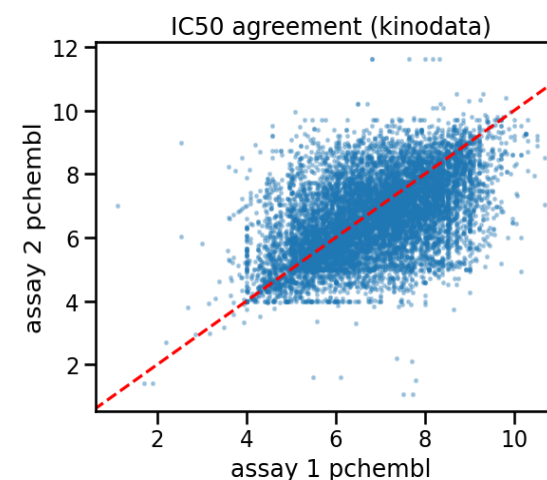
- Assays are inconsistent with each other
- Greg: *"Combining IC_{50} or K_i Values from Different Sources Is a Source of Significant Noise"*



$R^2 = 0.31$
 $MAE = 0.50$
Kendall's tau = 0.51



Landrum, Riniker., JCI, 2024, 64,5, <https://doi.org/10.1101/2024.09.10.612176>



$R^2=0.53$ $MAE=0.96$
Kendall's tau=0.37

- Pat Walters: *"Only use data from a single lab"* **Polaris**
- *Do we really want to throw away the massive amount of data?*

What we will do in detail ...

Session 1

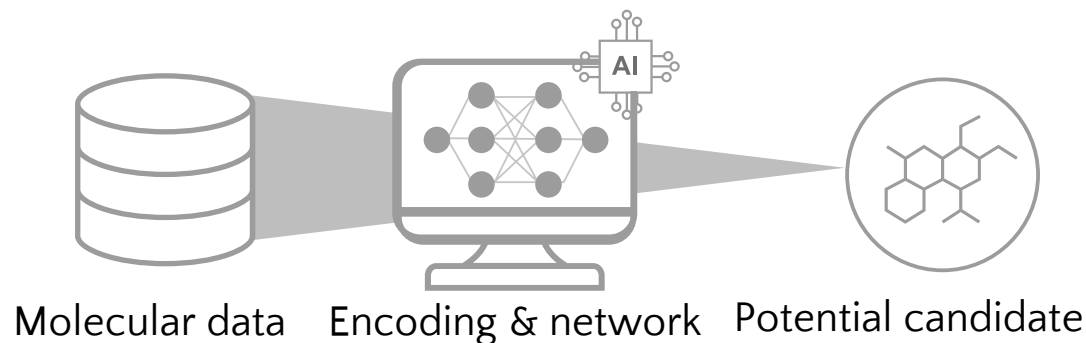
- CADD, Artificial Intelligence (AI) and Machine Learning (ML)
- The importance of data quality for AI/ML: Data preprocessing and analysis
- **Molecular representations: Fingerprints and similarities**

Session 2

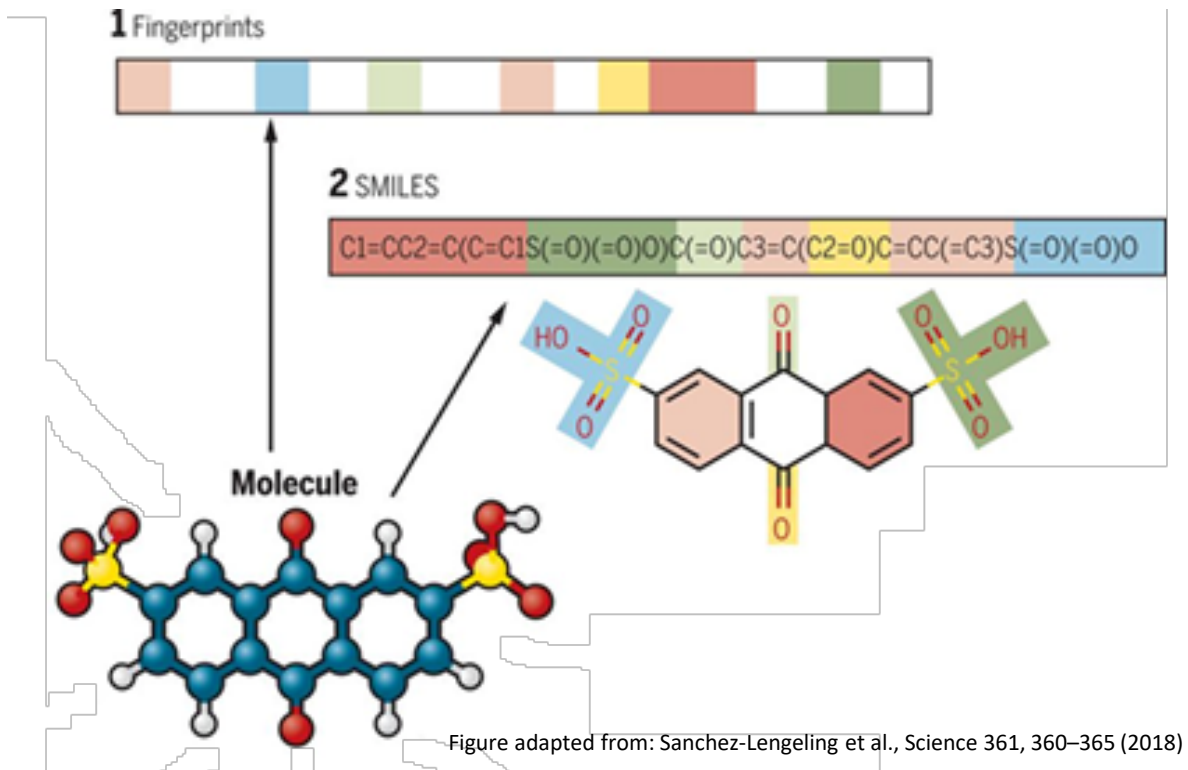
- Introduction machine learning (ML)
- (Un)supervised learning: RF, SVM, ...
- Performance evaluation

Session 3

- Deep learning
- Neuronal networks
- Molecular encodings and architectures



Representation and Encoding of Molecules



Working with molecules

Canonical
SMILES

Molecular graph

Fingerprints



Open-Source Cheminformatics
and Machine Learning

<https://www.rdkit.org/>

Representation and Encoding of Molecules

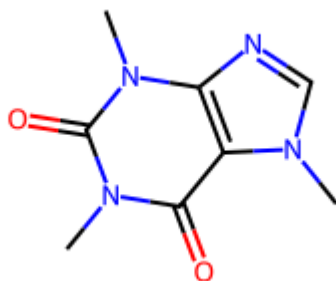
```
from rdkit import Chem
from rdkit.Chem import Draw
from rdkit.Chem import AllChem

# Input SMILES string
# Example: caffeine
smiles = "CN1C=NC2=C1C(=O)N(C(=O)N2C)C"

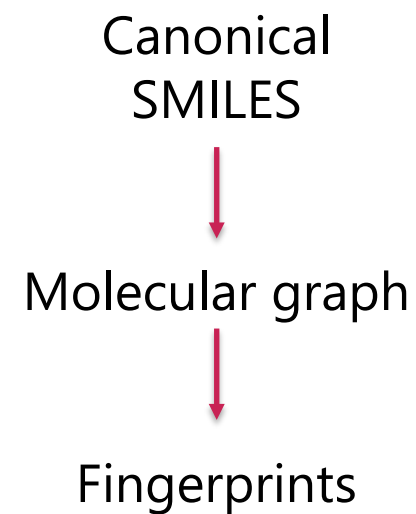
# Convert SMILES to molecule
mol = Chem.MolFromSmiles(smiles)

# Generate Morgan fingerprint
fingerprint = AllChem.GetMorganFingerprintAsBitVect(mol, radius=2, nBits=1024)

# Draw the molecule
mol
```



Working with molecules

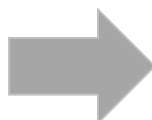
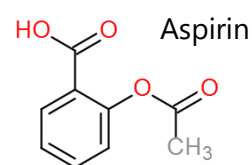


Open-Source Cheminformatics
and Machine Learning

<https://www.rdkit.org/>

Molecular Descriptors and Fingerprints

- Physico-chemical properties: molecular weight (MW), logP, PSA, etc.
 - Only **global** value for complete molecule



Feature	Value
MW	180.16
PSA	63.6
H-acceptor	4



Open-Source Cheminformatics
and Machine Learning

<https://www.rdkit.org/>

```
from rdkit.Chem import Descriptors
```

```
# Calculate descriptors
```

```
mw = Descriptors.MolWt(mol)
```

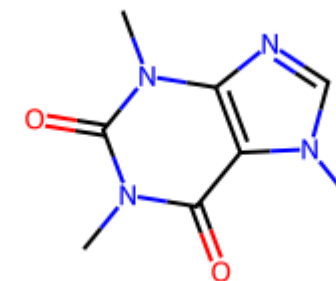
```
logp = Descriptors.MolLogP(mol)
```

```
print(f"Molecular Weight (MW): {mw:.2f}")
```

```
print(f"logP: {logp:.2f}")
```

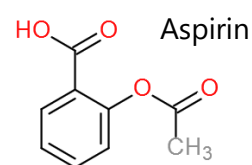
```
Molecular Weight (MW): 194.19
```

```
logP: -1.03
```



Molecular Descriptors and Fingerprints

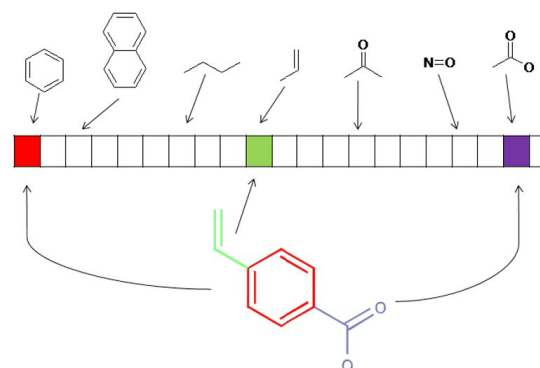
- Physico-chemical properties: molecular weight (MW), logP, PSA, etc.
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Feature	Value
MW	180.16
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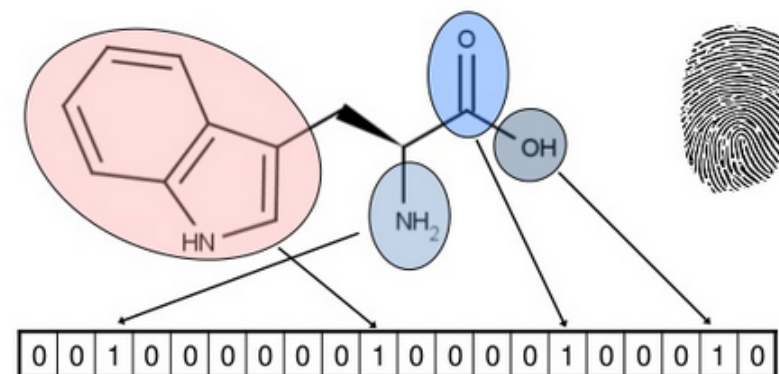
- ‚Molecular fingerprints‘: chemical structure and functional groups
 - Very good in distinguishing between active and inactive molecules

Molecular Fingerprint:
Each bit corresponds to a
predefined fragment of the
molecule



Transform molecule into a computer readable bit string

- Idea : Apply a transformation to a molecule to generate a bit vector or count vector (less frequent)
- Typical fingerprint algorithm
 - Extract features of the molecule
 - Hash the features (= convert them to a number in a unique way)
 - Use the hash to determine which bits should be set
- Molecular fingerprint sizes
 - Few hundred to several thousand bits



Fingerprints Based Predefined Substructural Keys MACCS (Molecular ACCess System)

Report the presence or absence of structural fragments

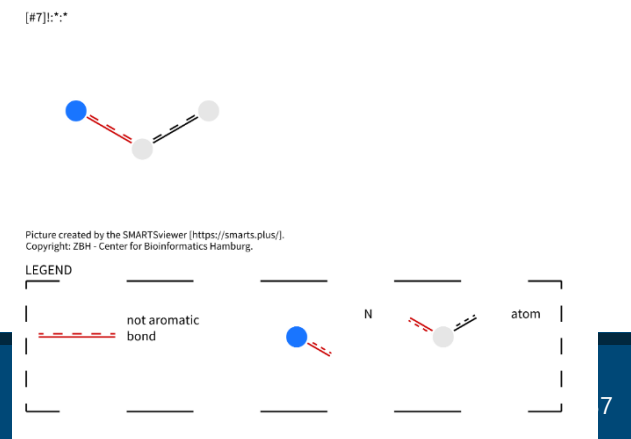
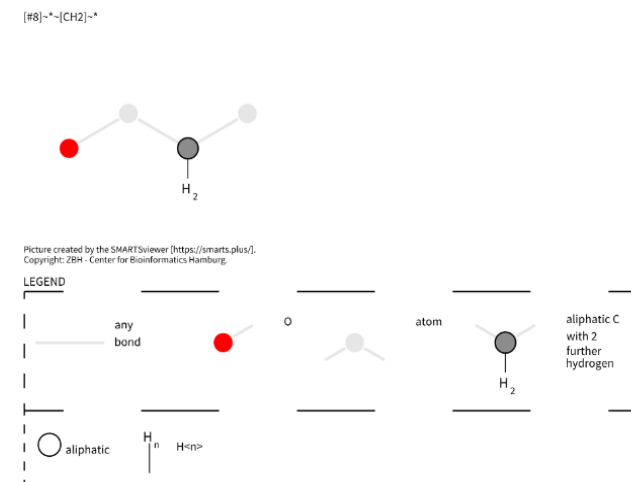
- 166 public keys
- Originally intended as a substructure fingerprint
- Each bit is associated with a SMARTS pattern
- Example bit definitions from the RDKit's implementation:
 - 132:('[#8]~*~[CH2]~*',0), # OACH2A
 - 133:('*@*!@[#7]',0), # A\$A!N
 - 134:('[F,Cl,Br,I]',0), # X (HALOGEN)
 - 135:('[#7]!~*~*',0), # Nnot%A%A
 - 136:('[#8]=*',1), # O=A>1

- **Results are interpretable**
- **Restricted to predefined number of rules**

J. L. Durant, et al., J. Chem. Inf. Comput. Sci. 2002, 42, 1273–1280.

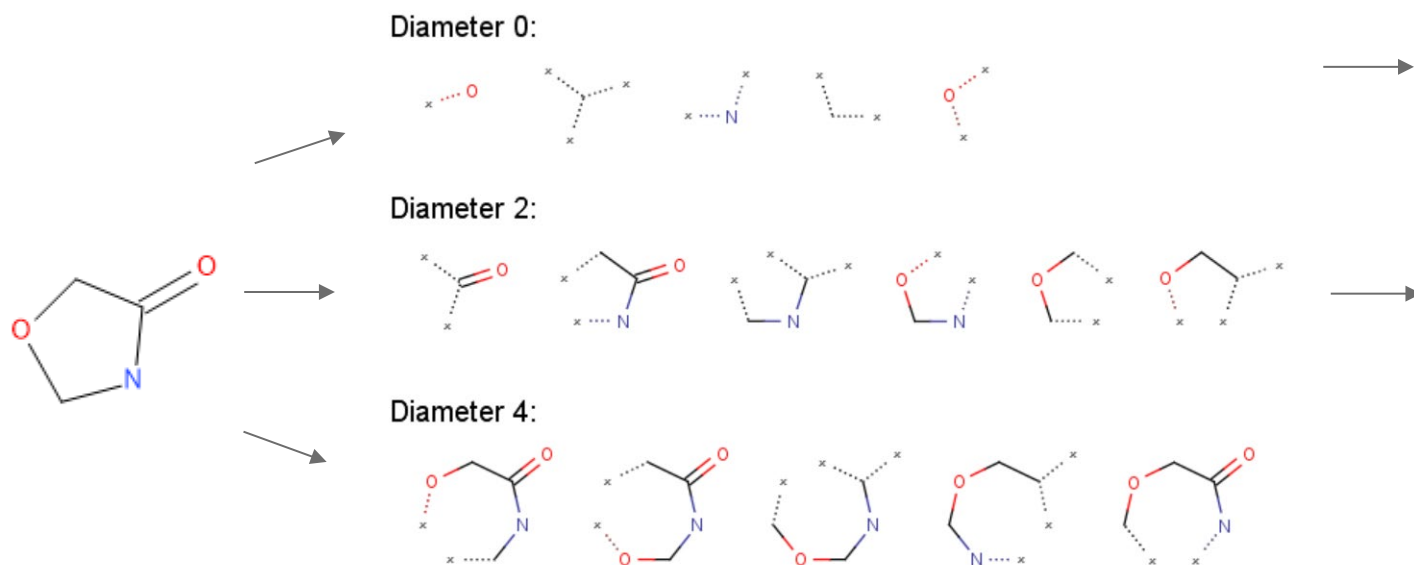
Check out SmartsViewer

<https://smarts.plus/>



Circular Fingerprints: ECFP/Morgan FP

"Circular fingerprint encodes fragments grown radially from each heavy atom up to a set radius, hashed into a fixed-length bitvector."



Identifiers

Identifier 1: -1266712900
Identifier 2: -1216914295
Identifier 3:

...

...

Main parameter:

- max diameter (default = 4)
- fingerprint length (default = 1024)
- counts of identifier (default = "No")

Identifier list:

Identifier 1 Identifier 2 Identifier 3 Identifier 4 Identifier 5 ...

Fixed-length binary representation

010000000100000111000000100000010000001100000001100

Hash function

Further information:

<https://docs.chemaxon.com/pages/viewpage.action?pageId=14483752>

Given a (set of) molecule(s) with known binding affinity to a specific target,
find further similar molecules

- Transform molecules into fingerprints
- Calculate similarity between each pair of fingerprints



A: 1 1 0 0 0 0 0 1 0 0 0



B: 1 1 0 0 0 0 1 1 0 1 0

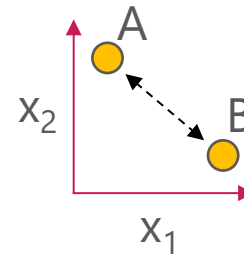


Similarity?

Fingerprint Comparison Metrics

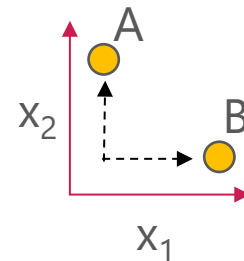
- Euclidean distance

$$D_{A,B} = \sqrt{\sum_i (x_{iA} - x_{iB})^2}$$



- Manhattan distance

$$D_{A,B} = \sum_i |x_{iA} - x_{iB}|$$



Consider both presence and absence of attributes

- Tanimoto

$$D_{A,B} = \frac{c}{a + b - c}$$

c = bits set in (A AND B)
 a = bits set in A
 b = bits set in B

Considers only presence of attributes

- Dice

$$D_{A,B} = \frac{c}{0.5 * (a + b)}$$

A: 1 1 0 0 0 0 0 1 0 0 0
B: 1 1 0 0 0 0 0 1 1 0 0
↳ $T = 3 / (5 + 3 - 3) = 0.6$

In need of ONE resource to get into python programming for CADD-related problems ?

UNIVERSITÄT
DES
SAARLANDES

What? Pipelines for common CADD* tasks using open resources

How? Coding-based Jupyter notebooks¹ (Python) & GUI-based KNIME workflows²



For what? Teaching, self studies & starting point for research projects

From and for whom? Students, group, community -> Beginners & advanced users

*CADD = computer-aided drug design

T002 · Molecular filtering: ADME and lead-likeness criteria

Authors:

- Michele Vichmann, CADD seminars 2017, Charité-FU Berlin
- Matthias Völzberg, CADD seminars 2018, Charité-FU Berlin
- Dominique Sycow, 2018-2020, [Volkamer Lab](#), Charité
- Andrea Volkamer, 2018-2020, [Volkamer Lab](#), Charité

Talktorial T002: This talktorial is part of the TeachOpenCADD pipeline described in the [first TeachOpenCADD paper](#), comprising of talktorials T001-T010.

ADME - absorption, distribution, metabolism, and excretion

Pharmacokinetics are mainly divided into four steps: Absorption, Distribution, Metabolism, and Excretion. These are summarized as **ADME**. Often, ADME also includes Toxicology and is thus referred to as ADMET or ADMETox. Below, the ADME steps are discussed in more detail ([Wikipedia](#)) and [Mol Pharm.](#) (2010). 7(5), 1388-1405).

Absorption: The amount and the time of drug-uptake into the body depends on multiple factors which can vary between individuals and their conditions as well as on the properties of the substance. Factors such as (poor) compound solubility, gastric emptying time, intestinal transit time, chemical (in-)stability in the stomach, and (in-)ability to permeate the intestinal wall can all influence the extent to which a drug is absorbed after e.g. oral administration, inhalation, or contact to skin.

Distribution: The distribution of an absorbed substance, i.e. within the body, between blood and different tissues, and crossing the blood-brain barrier are affected by regional blood flow rates, molecular size and polarity of the compound, and binding to serum proteins and transporter enzymes. Critical effects in toxicology can be the accumulation of highly apolar substances in fatty tissue, or crossing of the blood-brain barrier.

Metabolism: After entering the body, the compound will be metabolized. This means that only part of this compound will actually reach its target. Mainly liver and kidney enzymes are responsible for the break-down of xenobiotics (substances that are extrinsic to the body).

Excretion: Compounds and their metabolites need to be removed from the body via excretion, usually through the kidneys (urine) or in the feces. Incomplete excretion can result in accumulation of foreign substances or adverse interference with normal metabolism.

Figure 1: ADME processes in the human body (figure taken from [Openclipart](#) and adapted).

Discussion

In this talktorial, we have learned about Lipinski's Ro5 as a measure to estimate a compound's oral bioavailability and we have applied the rule on a dataset using rdkit. Note that drugs can also be administered via alternative routes, i.e. inhalation, skin penetration and injection.

In this talktorial, we have looked at only one of many more ADME properties. Webserver such as [SwissADME](#) give a more comprehensive view on compound properties.

Quiz

- In what way can the chemical properties described by the Ro5 affect ADME?
- Find or design a molecule which violates three or four rules.
- How can you plot information for an additional molecule in the radar charts that we have created in this talktorial?

Aim of this talktorial

In the context of drug design, it is important to filter candidate molecules by e.g. their physicochemical properties. In this talktorial, the compounds acquired from ChEMBL (**Talktorial 001**) will be filtered by Lipinski's rule of five to keep only orally bioavailable compounds.

Contents in Theory

- ADME - absorption, distribution, metabolism, and excretion
- Lead-likeness and Lipinski's rule of five (Ro5)
- Radar charts in the context of lead-likeness

Contents in Practical

- Define and visualize example molecules
- Calculate and plot molecular properties for Ro5
- Investigate compliance with Ro5
- Apply Ro5 to the EGFR dataset
- Visualize Ro5 properties (radar plot)

References

- ADME criteria ([Wikipedia](#) and [Mol Pharm.](#) (2010). 7(5), 1388-1405)
- [SwissADME](#) webserver
- What are lead compounds? ([Wikipedia](#))
- What is the LogP value? ([Wikipedia](#))
- Lipinski et al. "Experimental and computational approaches to estimate solubility and permeability in drug discovery and development settings." (*Adv. Drug Deliv. Rev.* (1997). 23, 3-25)
- Ritzke et al. "Graphical representation of ADME-related molecule properties for medicinal chemists" (*Drug Discov. Today* (2011). 16, 65-72)

```
In [16]: molecules["molecular_weight"] = molecules["ROMol"].apply(descriptors.ExactMolwt)  
molecules["n_hba"] = molecules["ROMol"].apply(descriptors.NumHAcceptors)  
molecules["n_hbd"] = molecules["ROMol"].apply(descriptors.NumHDonors)  
molecules["logp"] = molecules["ROMol"].apply(descriptors.MolLogp)  
# Colors are used for plotting the molecules later  
molecules["color"] = ["red", "green", "blue", "cyan"]  
# NBVAL_CHECK_OUTPUT  
molecules[["molecular_weight", "n_hba", "n_hbd", "logp"]]
```

	molecular_weight	n_hba	n_hbd	logp
0	1201.841368	12	5	3.26900
1	306.184447	4	1	1.68492
2	536.438202	0	0	12.60580
3	314.224580	2	2	5.84650

```
In [8]: # full preview  
molecules
```

out[8]:	name	smiles	ROMol	molecular_weight	n_hba	n_hbd	lc
0	cyclosporine	CCC1C(O)N(CC(C=O)N(CC(C=O)NC(C=O)N(C(C=O)NC(...		1201.841368	12	5	3.
1	diazepam	CN1CC(NCC1)C2=C3C=CC(=O)C=C3N4=C(C(N2)C=C(C4)C		306.184447	4	1	1.
2	beta-carotene	CC1=C(C(CCC1)(C)C)=CC(=C)C=CC=CC=C(C)C=CC=...		536.438202	0	0	1;
3	cannabidiol	CCCCCCC1=CC(C=C1)OC2C=C(CCC2C(C)C)CO		314.224580	2	2	5.

Explanatory text
Executable code
Code output

All-in-one Jupyter Notebook

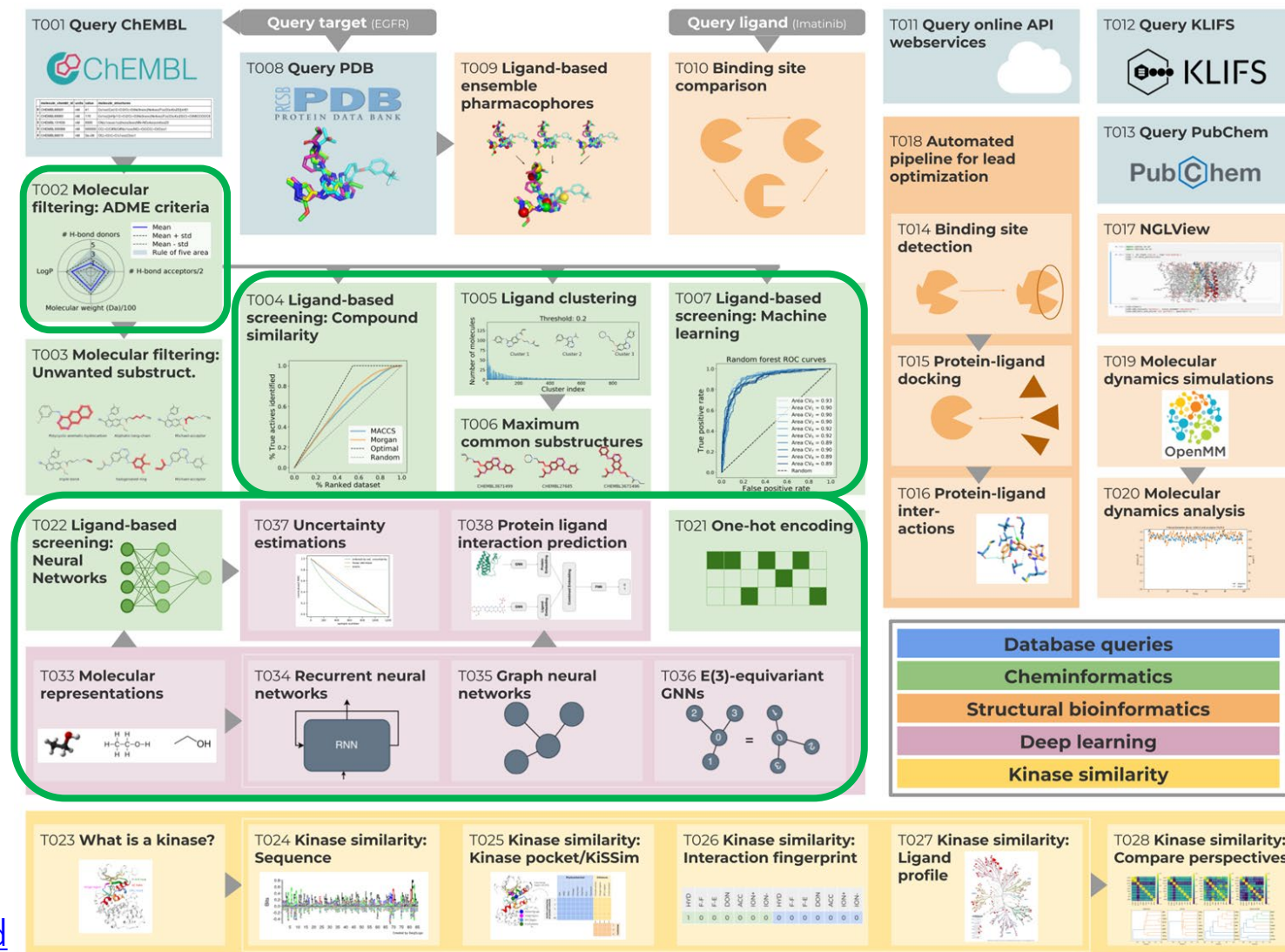
jupyter

TeachOpenCADD – Open-Source Python Pipelines

Relevant for today's sessions

Cheminformatics
and ML examples

DL examples



<https://github.com/volkamerlab/teachopencadd>

Hands on ...

Time	Topic	Presenter
08:30-09:10	Lecture: From molecular data to models	Andrea Volkamer
09:10-10:00	Hands-on: Data	Katharina Buchthal
10:00-10:40	Lecture: Machine learning (ML) essentials	Andrea Volkamer
10:40-11:30	Hands-on: ML	Katharina Buchthal
11:30-12:30	Break	
12:30-13:00	Hands-on: continued	Katharina Buchthal
13:00-13:40	Lecture: Deep learning (DL) applications	Andrea Volkamer
13:40-14:30	Hands-on: DL	Katharina Buchthal

https://github.com/volkamerlab/cic_summerschool_2025